

Multiple Choice (10 marks)

1. Adding more basis functions in a linear model, pick the most probably option:
 - (a) Decreases model bias
 - (b) Decreases estimation bias
 - (c) Decreases variance
 - (d) Doesn't affect bias and variance
2. Statement 1| The softmax function is commonly used in multiclass logistic regression.
Statement 2| The temperature of a nonuniform softmax distribution affects its entropy.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Suppose your model is overfitting. Which of the following is NOT a valid way to try and reduce the overfitting?
 - (a) Increase the amount of training data.
 - (b) Improve the optimisation algorithm being used for error minimisation.
 - (c) Decrease the model complexity.
 - (d) Reduce the noise in the training data.
4. Existential risks posed by AI are most commonly associated with which of the following professors?
 - (a) Nando de Freitas
 - (b) Yann LeCun
 - (c) Stuart Russell
 - (d) Jitendra Malik
5. _ refers to a model that can neither model the training data nor generalize to new data.
 - (a) good fitting
 - (b) overfitting
 - (c) underfitting
 - (d) all of the above

True / False (10 marks)

Answer True or False

6. True or False: The numerical output of a sigmoid node in a neural network is bounded between 0 and 1.
7. True or False: Out-of-distribution detection is synonymous with background detection in machine learning applications.
8. True or False: Highway networks were developed before ResNets and use max pooling instead of convolutions.
9. True or False: The dimensionality of the null space of the matrix A is 2.
10. True or False: A larger number of test examples is needed to achieve statistically significant results when the error rate is smaller.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to count the elements in a list until an element is a tuple.
12. Write a function to perform the rear element extraction from list of tuples records.

End of Paper